

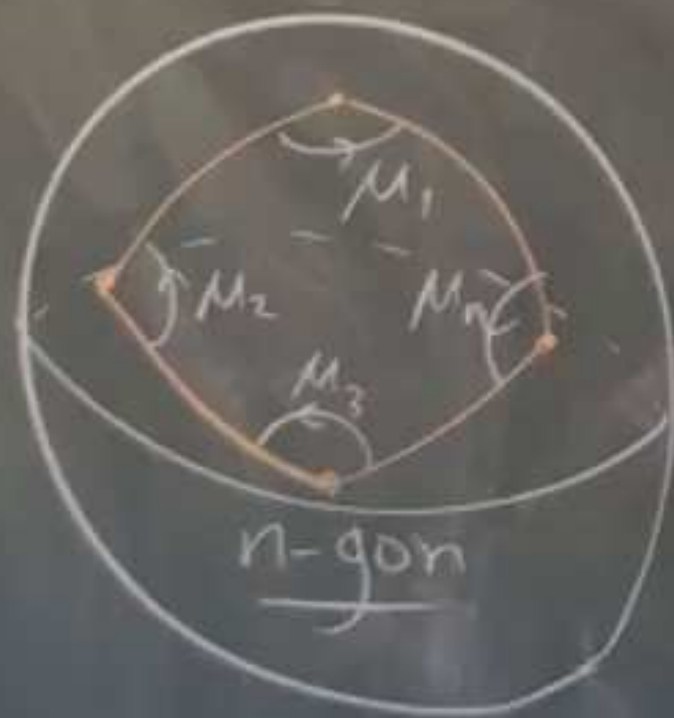
Algebraic Topology: a beginner's course

This lecture:

**AlgTop16: Rational curvature
of polygons and Euler's formula**

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Lecture 16: Rational curvature of polytopes and the Euler number



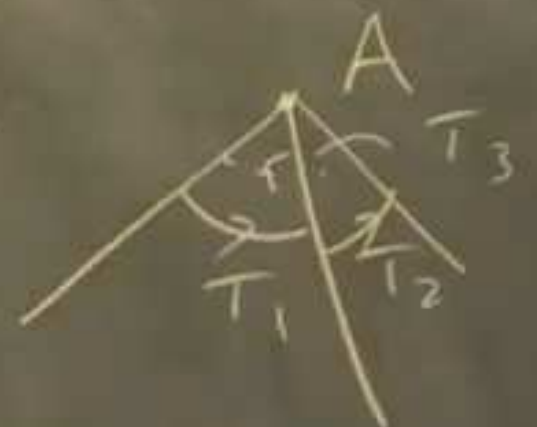
Harmon's thm

w/ (strictly) concave n -gon)

$$+ \mu_n - \frac{n}{2} + 1$$

types

Ex. A has degree 3,

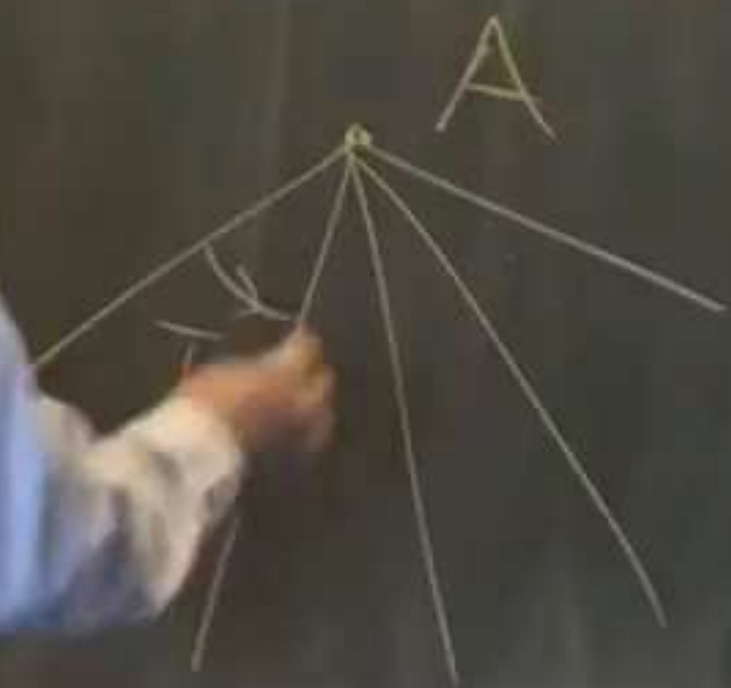


$$w = \mu_1 + \mu_2 + \mu_3 - \frac{1}{2}$$

$$= \left(\frac{1}{2} - T_1\right) + \left(\frac{1}{2} - T_2\right) + \left(\frac{1}{2} - T_3\right) - \frac{1}{2}$$

$$= 1 - (T_1 + T_2 + T_3)$$

Ex. A



Thm If a polytope P has vertex A
with tangents of faces at A being $T_1, T_2, \dots, T_n$,
then curvature (P) at $A = 1 - (T_1 + T_2 + \dots + T_n)$

Prob. 18 Compute directly the curvature
at a vertex for a i) truncated octahedron,
ii) truncated icosahedron. Find total
curvature.

Ex. 2

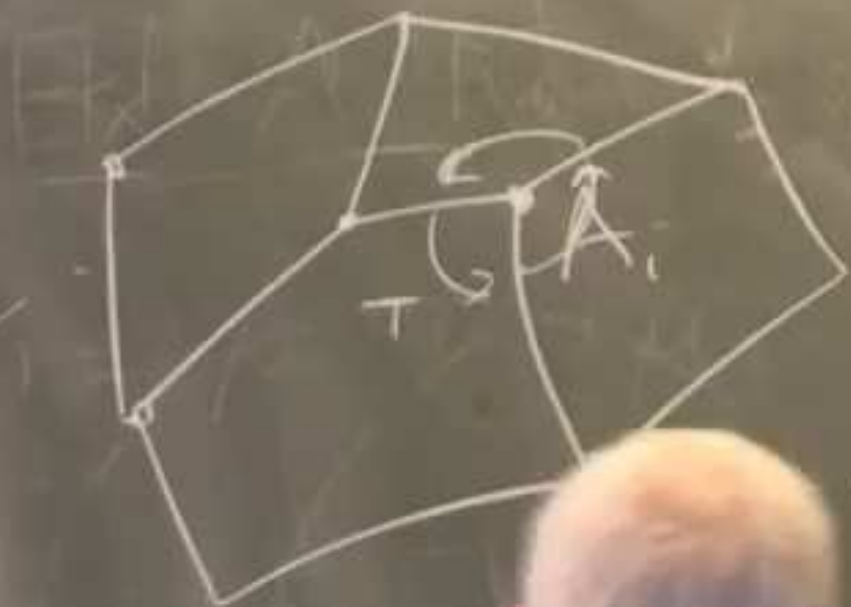
Ex. 3

Prob. 18 Compute directly the curvature at a vertex for a i) truncated octahedron, ii) truncated icosahedron. Find total curvatures.

Suppose we have a number of polygonal faces, meeting two at a time along an edge, to form a polyhedron.

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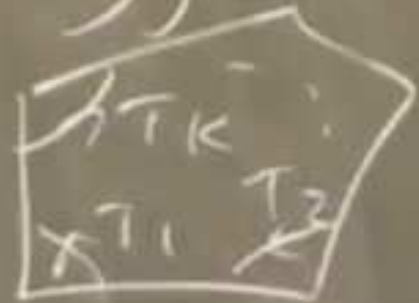
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Total curvature = $\sum_{\text{vertices } A} \text{curv}(A) =$

planar polygon.

$$T_1 + \dots + T_k = \frac{k}{2} - 1$$



$$\sum_{\text{vertices } A_i} \text{curv}(A_i) = \sum_{\text{vertices } A_i} \left[1 - (T_1 + \dots + T_k) \right]$$

$$= V - \sum_{\text{faces}} (\text{sum of triangles})$$

$$= V - \sum_{\text{faces}} \left(\frac{k_i}{2} - 1 \right) = V - E + F = \chi //$$

Prob 18 Compute directly the curvature
at a vertex for a i) truncated octahedron,
ii) truncated icosahedron. Find total
curvature



Total curvature

$$\text{cur}(A_i) =$$